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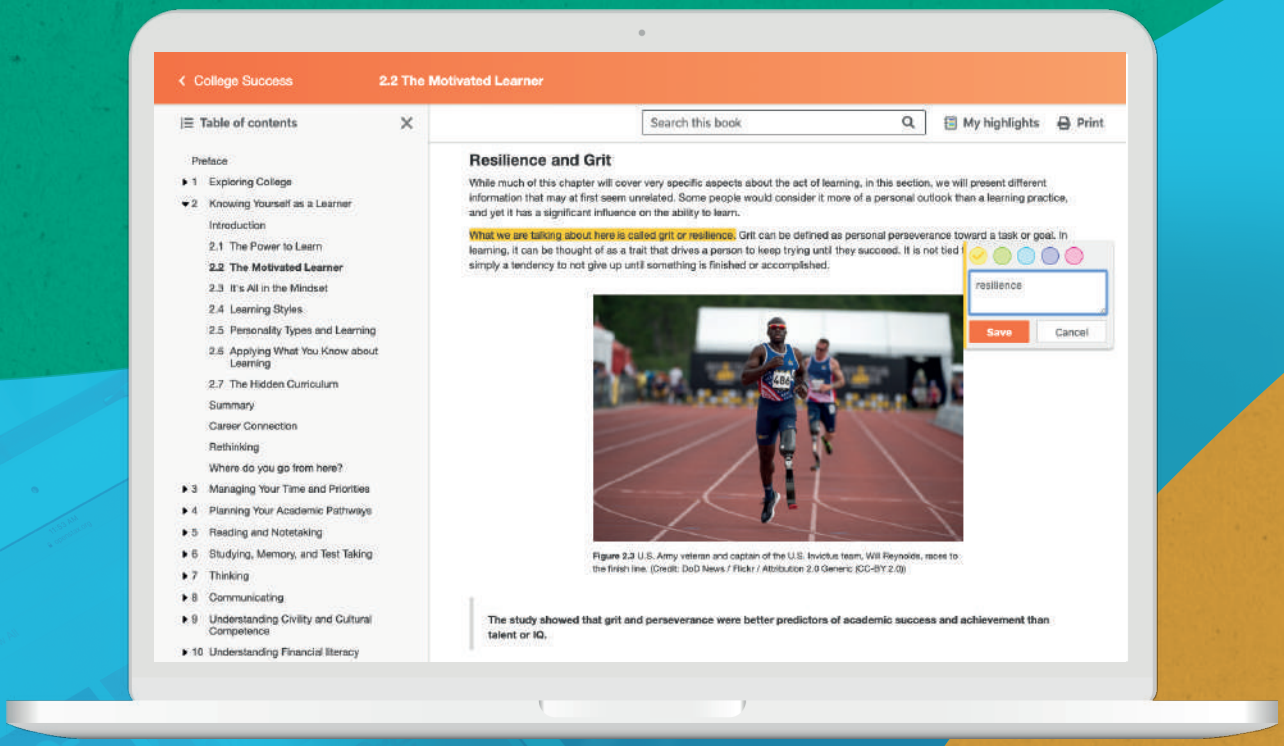

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Preface

Welcome to *Psychology 2e*, an OpenStax resource. This textbook was written to increase student access to high-quality learning materials, maintaining highest standards of academic rigor at little to no cost.

About OpenStax

OpenStax is a nonprofit based at Rice University, and it's our mission to improve student access to education. Our first openly licensed college textbook was published in 2012, and our library has since scaled to over 35 books for college and AP® courses used by hundreds of thousands of students. OpenStax Tutor, our low-cost personalized learning tool, is being piloted in college courses throughout the country. Through our partnerships with philanthropic foundations and our alliance with other educational resource organizations, OpenStax is breaking down the most common barriers to learning and empowering students and instructors to succeed.

About OpenStax Resources

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All OpenStax textbooks undergo a rigorous review process. However, like any professional-grade textbook, errors sometimes occur. Since our books are web based, we can make updates periodically when deemed pedagogically necessary. If you have a correction to suggest, submit it through the link on your book page on openstax.org. Subject matter experts review all errata suggestions. OpenStax is committed to remaining transparent about all updates, so you will also find a list of past errata changes on your book page on openstax.org.

Format

You can access this textbook for free in web view or PDF through openstax.org, and for a low cost in print.

About *Psychology 2e*

Psychology 2e is designed to meet scope and sequence requirements for the single-semester introduction to psychology course. The book offers a comprehensive treatment of core concepts, grounded in both classic

studies and current and emerging research. The text also includes coverage of the DSM-5 in examinations of psychological disorders. *Psychology 2e* incorporates discussions that reflect the diversity within the discipline, as well as the diversity of cultures and communities across the globe.

Coverage and scope

The first edition of *Psychology* has been used by thousands of faculty and hundreds of thousands of students since its publication in 2015. OpenStax mined our adopters' extensive and helpful feedback to identify the most significant revision needs while maintaining the organization that many instructors had incorporated into their courses. Specific surveys, pre-revision reviews, and customization analysis, as well as analytical data from OpenStax partners and online learning environments, all aided in planning the revision.

The result is a book that thoroughly treats psychology's foundational concepts while adding current and meaningful coverage in specific areas. *Psychology 2e* retains its manageable scope and contains ample features to draw learners into the discipline.

Structurally, the textbook remains similar to the first edition, with no chapter reorganization and very targeted changes at the section level.

- Chapter 1: Introduction to Psychology
- Chapter 2: Psychological Research
- Chapter 3: Biopsychology
- Chapter 4: States of Consciousness
- Chapter 5: Sensation and Perception
- Chapter 6: Learning
- Chapter 7: Thinking and Intelligence
- Chapter 8: Memory
- Chapter 9: Lifespan Development
- Chapter 10: Motivation and Emotion
- Chapter 11: Personality
- Chapter 12: Social Psychology
- Chapter 13: Industrial-Organizational Psychology
- Chapter 14: Stress, Lifestyle, and Health
- Chapter 15: Psychological Disorders
- Chapter 16: Therapy and Treatment

Changes to the Second Edition

OpenStax only undertakes second editions when significant modifications to the text are necessary. In the case of *Psychology 2e*, user feedback indicated that we needed to focus on a few key areas, which we have done in the following ways.

Content revisions for clarity, accuracy, and currency

The revision plan varied by chapter based on need. Some chapters were significantly updated for conceptual coverage, research-informed data, and clearer language. In other chapters, the revisions focused mostly on currency of examples and updates to statistics.

Over 210 new research references have been added or updated in order to improve the scholarly underpinnings of the material and broaden the perspective for students. Dozens of examples and feature boxes have been changed or added to better explain concepts and/or increase relevance for students.

Research replication and validity

To engage students in stronger critical analysis and inform them about research reproducibility, substantial coverage has been added to the research chapter and strategically throughout the textbook whenever key

studies are discussed. This material is presented in a balanced way and provides instructors with ample opportunity to discuss the importance of replication in a manner that best suits their course.

Diversity, representation, and inclusion

With the help of researchers and teachers who focus on diversity- and identity-related issues, OpenStax has engaged in detailed diversity reviews to identify opportunities to improve the textbook. Reviewers were asked to follow a framework to evaluate the book's terminology, research citations, key contributors to the field, photos and illustrations, and related aspects, commenting on the representation and consideration of diverse groups. Significant additions and revisions were made in this regard, and the review framework itself is available among the OpenStax *Psychology 2e* instructor resources.

Art and illustrations

Under the guidance of the authors and expert scientific illustrators, especially those well versed in creating accessible art, the OpenStax team made changes throughout the art program in *Psychology 2e*.

Accessibility improvements

As with all OpenStax books, the first edition of *Psychology* was created with a focus on accessibility. We have emphasized and improved that approach in the second edition. Our goal is to ensure that all OpenStax websites and the web view versions of our learning materials follow accessible web design best practices, so that they will meet the W3C-WAI Web Content Accessibility Guidelines (WCAG) 2.0 at Level AA and Section 508 of the Rehabilitation Act. The WCAG 2.0 guidelines explain ways to make web content more accessible for people with disabilities and more user-friendly for everyone.

- To accommodate users of specific assistive technologies, all alternative text was reviewed and revised for comprehensiveness and clarity.
- All illustrations were revised to improve the color contrast, which is important for some visually impaired students.
- Overall, the OpenStax platform has been continually upgraded to improve accessibility.

To learn more about our commitment and progress, please view our accessibility statement (https://openstax.org/accessibility-statement).

A transition guide will be available on openstax.org to highlight the specific chapter-level changes to the second edition.

Pedagogical foundation

Psychology 2e engages students through inquiry, self-reflection, and investigation. Features in the second edition have been carefully updated to remain topical and relevant while deepening students' relationship to the material. They include the following:

- **Everyday Connection** features tie psychological topics to everyday issues and behaviors that students encounter in their lives and the world. Topics include the validity of scores on college entrance exams, the opioid crisis, the impact of social status on stress and healthcare, and cognitive mapping.
- **What Do You Think?** features provide research-based information and ask students their views on controversial issues. Topics include "Brain Dead and on Life Support," "Violent Media and Aggression," and "Capital Punishment and Criminals with Intellectual Disabilities."
- **Dig Deeper** features discuss one specific aspect of a topic in greater depth so students can dig more deeply into the concept. Examples include discussions on the distinction between evolutionary psychology and behavioral genetics, recent findings on neuroplasticity, the field of forensic psychology, and a presentation of research on strategies for coping with prejudice and discrimination.
- **Connect the Concepts** features revisit a concept learned in another chapter, expanding upon it within a different context. Features include "Emotional Expression and Emotional Regulation," "Tweens, Teens,

and Social Norms," and "Conditioning and OCD."

Art, interactives, and assessments that engage

Our art program is designed to enhance students' understanding of psychological concepts through simple, effective graphs, diagrams, and photographs. *Psychology 2e* also incorporates links to relevant interactive exercises and animations that help bring topics to life. Selected assessment items touch directly on students' lives.

- **Link to Learning** features direct students to online interactive exercises and animations that add a fuller context to core content and provide an opportunity for application.
- **Personal Application Questions** engage students in topics at a personal level to encourage reflection and promote discussion.

Additional Resources

Student and Instructor Resources

We've compiled additional resources for both students and instructors, including Getting Started Guides, an instructor solution guide, a test bank, and PowerPoint slides. Instructor resources require a verified instructor account, which you can apply for when you log in or create your account on openstax.org. Take advantage of these resources to supplement your OpenStax book.

Community Hubs

OpenStax partners with the Institute for the Study of Knowledge Management in Education (ISKME) to offer Community Hubs on OER Commons—a platform for instructors to share community-created resources that support OpenStax books, free of charge. Through our Community Hubs, instructors can upload their own materials or download resources to use in their own courses, including additional ancillaries, teaching material, multimedia, and relevant course content. We encourage instructors to join the hubs for the subjects most relevant to your teaching and research as an opportunity both to enrich your courses and to engage with other faculty.

To reach the Community Hubs, visit www.oercommons.org/hubs/openstax.

Technology partners

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About the authors

Senior contributing authors

Rose M. Spielman (Content Lead)
Dr. Rose Spielman has been teaching psychology and working as a licensed clinical psychologist for 20 years. Her academic career has included positions at Quinnipiac University, Housatonic Community College, and Goodwin College. As a licensed clinical psychologist, educator, and volunteer director, Rose is able to connect with people from diverse backgrounds and facilitate treatment, advocacy, and education. In her years of work as a teacher, therapist, and administrator, she has helped thousands of students and clients and taught them to advocate for themselves and move their lives forward to become more productive citizens and family members.

William J. Jenkins, Mercer University
Marilyn D. Lovett, Spelman College

Contributing Authors

Mara Aruguete, Lincoln University
Laura Bryant, Eastern Gateway Community College
Barbara Chappell, Walden University
Kathryn Dumper, Bainbridge State College
Arlene Lacombe, Saint Joseph's University
Julie Lazzara, Paradise Valley Community College
Tammy McClain, West Liberty University
Barbara B. Oswald, Miami University
Marion Perlmutter, University of Michigan
Mark D. Thomas, Albany State University

Reviewers

Patricia G. Adams, Pitt Community College
Daniel Bellack, Trident Technical College
Christopher M. Bloom, Providence College
Jerimy Blowers, Cayuga Community College
Salena Brody, Collin College
David A. Caicedo, Borough of Manhattan Community College, CUNY
Bettina Casad, University of Missouri–St. Louis
Sharon Chacon, Northeast Wisconsin Technical College
James Corpening
Frank Eyetsemitan, Roger Williams University
Tamara Ferguson, Utah State University
Kathleen Flannery, Saint Anselm College
Johnathan Forbey, Ball State University
Laura Gaudet, Chadron State College
William Goggin, University of Southern Mississippi
Jeffery K. Gray, Charleston Southern University
Heather Griffiths, Fayetteville State University
Mark Holder, University of British Columbia
Rita Houge, Des Moines Area Community College
Colette Jacquot, Strayer University
John Johanson, Winona State University
Andrew Johnson, Park University
Shaila Khan, Tougaloo College
Cynthia Kreutzer, Georgia State University Perimeter College at Clarkston Campus
Carol Laman, Houston Community College
Dana C. Leighton, Texas A&M University—Texarkana
Thomas Malloy, Rhode Island College
Jan Mendoza, Golden West College
Christopher Miller, University of Minnesota
Lisa Moeller, Beckfield College
Amy T. Nusbaum, Heritage University
Jody Resko, Queensborough Community College (CUNY)
Hugh Riley, Baylor University
Juan Salinas, University of Texas at Austin
Brittney Schrick, Southern Arkansas University
Phoebe Scotland, College of the Rockies

Christine Selby, Husson University
Sally B. Seraphin, Centre College
Brian Sexton, Kean University
Nancy Simpson, Trident Technical College
Jason M. Smith, Federal Bureau of Prisons – FCC Hazelton
Robert Stennett, University of Georgia
Jennifer Stevenson, Ursinus College
Eric Weiser, Curry College
Jay L. Wenger, Harrisburg Area Community College
Alan Whitehead, Southern Virginia University
Valjean Whitlow, American Public University
Rachel Wu, University of California, Riverside
Alexandra Zelin, University of Tennessee at Chattanooga

Introduction to Psychology 1

FIGURE 1.1 Psychology is the scientific study of mind and behavior. (credit "background": modification of work by Nattachai Noogure; credit "top left": modification of work by Peter Shanks; credit "top middle": modification of work by "devinf"/Flickr; credit "top right": modification of work by Alejandra Quintero Sinisterra; credit "bottom left": modification of work by Gabriel Rocha; credit "bottom middle-left": modification of work by Caleb Roenigk; credit "bottom middle-right": modification of work by Staffan Scherz; credit "bottom right": modification of work by Czech Provincial Reconstruction Team)

CHAPTER OUTLINE

1.1 What Is Psychology?
1.2 History of Psychology
1.3 Contemporary Psychology
1.4 Careers in Psychology

INTRODUCTION Clive Wearing is an accomplished musician who lost his ability to form new memories when he became sick at the age of 46. While he can remember how to play the piano perfectly, he cannot remember what he ate for breakfast just an hour ago (Sacks, 2007). James Wannerton experiences a taste sensation that is associated with the sound of words. His former girlfriend's name tastes like rhubarb (Mundasad, 2013). John Nash is a brilliant mathematician and Nobel Prize winner. However, while he was a professor at MIT, he would tell people that the *New York Times* contained coded messages from extraterrestrial beings that were intended for him. He also began to hear voices and became suspicious of the people around him. Soon thereafter, Nash was diagnosed with schizophrenia and admitted to a state-run mental institution (O'Connor & Robertson,

2002). Nash was the subject of the 2001 movie *A Beautiful Mind*. Why did these people have these experiences? How does the human brain work? And what is the connection between the brain's internal processes and people's external behaviors? This textbook will introduce you to various ways that the field of psychology has explored these questions.

1.1 What Is Psychology?

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Define psychology
- Understand the merits of an education in psychology

What is creativity? What are prejudice and discrimination? What is consciousness? The field of psychology explores questions like these. **Psychology** refers to the scientific study of the mind and behavior. Psychologists use the scientific method to acquire knowledge. To apply the scientific method, a researcher with a question about how or why something happens will propose a tentative explanation, called a hypothesis, to explain the phenomenon. A hypothesis should fit into the context of a scientific theory, which is a broad explanation or group of explanations for some aspect of the natural world that is consistently supported by evidence over time. A theory is the best understanding we have of that part of the natural world. The researcher then makes observations or carries out an experiment to test the validity of the hypothesis. Those results are then published or presented at research conferences so that others can replicate or build on the results.

Scientists test that which is perceivable and measurable. For example, the hypothesis that a bird sings because it is happy is not a hypothesis that can be tested since we have no way to measure the happiness of a bird. We must ask a different question, perhaps about the brain state of the bird, since this can be measured. However, we can ask individuals about whether they sing because they are happy since they are able to tell us. Thus, psychological science is empirical, based on measurable data.

In general, science deals only with matter and energy, that is, those things that can be measured, and it cannot arrive at knowledge about values and morality. This is one reason why our scientific understanding of the mind is so limited, since thoughts, at least as we experience them, are neither matter nor energy. The scientific method is also a form of empiricism. An **empirical method** for acquiring knowledge is one based on observation, including experimentation, rather than a method based only on forms of logical argument or previous authorities.

It was not until the late 1800s that psychology became accepted as its own academic discipline. Before this time, the workings of the mind were considered under the auspices of philosophy. Given that any behavior is, at its roots, biological, some areas of psychology take on aspects of a natural science like biology. No biological organism exists in isolation, and our behavior is influenced by our interactions with others. Therefore, psychology is also a social science.

WHY STUDY PSYCHOLOGY?

Often, students take their first psychology course because they are interested in helping others and want to learn more about themselves and why they act the way they do. Sometimes, students take a psychology course because it either satisfies a general education requirement or is required for a program of study such as nursing or pre-med. Many of these students develop such an interest in the area that they go on to declare psychology as their major. As a result, psychology is one of the most popular majors on college campuses across the United States (Johnson & Lubin, 2011). A number of well-known individuals were psychology majors. Just a few famous names on this list are Facebook's creator Mark Zuckerberg, television personality and political satirist Jon Stewart, actress Natalie Portman, and filmmaker Wes Craven (Halonen, 2011). About 6 percent of all bachelor degrees granted in the United States are in the discipline of psychology (U.S. Department of Education, 2016).